Trigonometry is basically the study of right triangles. It has 3 main functions: Sine (sin), Cosine (cos), and Tangent (tan). Each has a ratio to solve for missing angles and sides given some information.

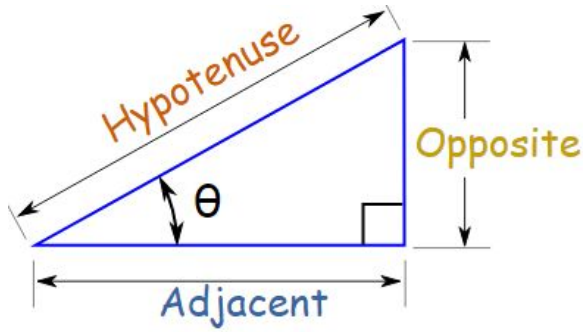
Here is a helpful mnemonic to help us remember the ratios for sin, cos, and tan.

**SOH CAH TOA**

SOH: $\sin(\theta) = \dfrac{opposite}{hypotenuse}$

CAH: $\cos(\theta) = \dfrac{adjacent}{hypotenuse}$

TOA: $\tan(\theta) = \dfrac{opposite}{adjacent}$



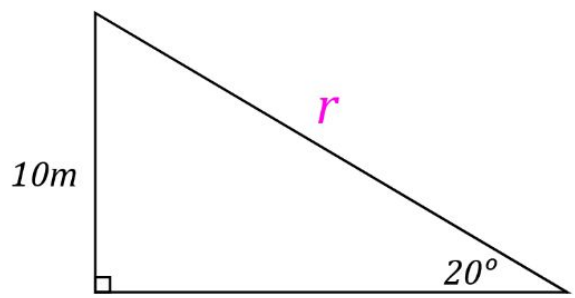

The "opposite" side is the triangle side *opposite* of the angle ( $\theta$ ) in the trigonometric function.

The "adjacent" side is the triangle side next to the angle ( $\theta$ ) in the trigonometric function.

While, the "hypotenuse" is always the longest triangle side.

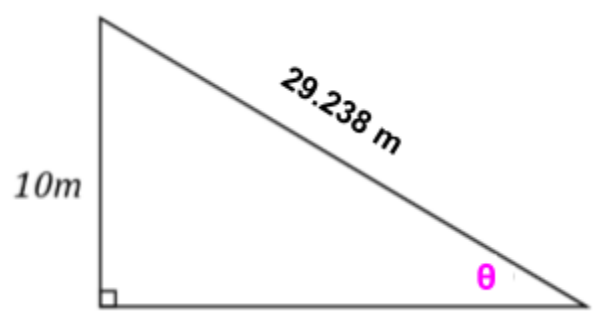
**Finding the missing side length using SOH CAH TOA**

1. Look at the picture on the right. To find "r", we must first recognize that side "r" is the hypotenuse of the triangle and the side of 10m is opposite of the 20° angle that is given.
2. Using SOH, we get $\sin(20°) = \frac{10}{r}$
3. Rewriting the equation to solve for r, we get r = $\frac{10}{\sin(20°)}$
4. Using our calculator r = 29.238m.



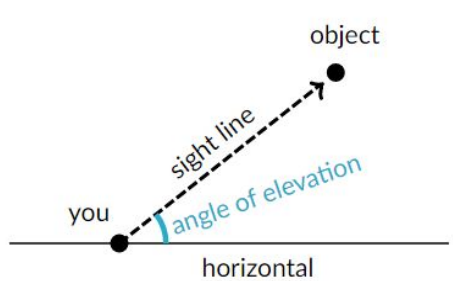

**Finding an angle using SOH CAH TOA (solving backward)**

1. Using SOH, we get $\sin(\theta) = \frac{10}{29.238} = .342$
2. To solve for θ, we need to use inverse sin ( $\sin^{-1}$ (_) ) and rewrite the equation into $\theta = \sin^{-1}(.342)$
3. Putting $\sin^{-1}(.342)$ into the calculator, we get θ = 20 ° (the same angle we started with)

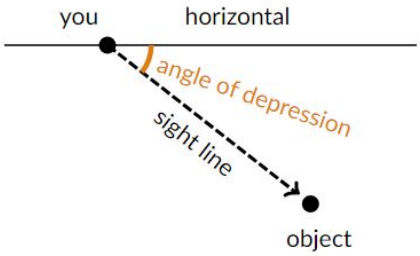



**Solving angle of elevations and depressions**

Angle of elevation is when an object is above the level of the observer. It's the angle between the observer's horizontal line of sight and their line of sight up to the object.

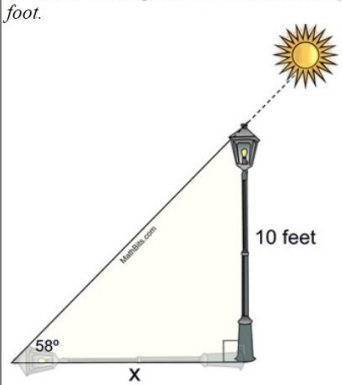


Angle of depression is when an object is below the level of the observer. It's the angle between the observer's horizontal line of sight and their line of sight down to the object.

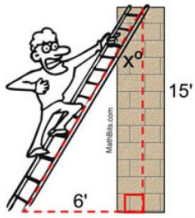


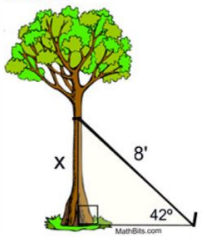

**Practice problems**

Find the shadow cast by a 10 foot lamp post when the angle of elevation of the sun is 58°. Find the length to the *nearest tenth of a foot.*




A ladder leans against a brick wall. The foot of the ladder is 6 feet from the wall. The ladder reaches a height of 15 feet on the wall. Find to the *nearest degree*, the angle the ladder makes with the wall.




https://mathbitsnotebook.com/Geometry/Trigonometry/TGElevDepress.html

A nursery plants a new tree and attaches a guy wire to help support the tree while its roots take hold. An eight foot wire is attached to the tree and to a stake in the ground. From the stake in the ground the angle of elevation of the connection with the tree is 42°. Find to the *nearest tenth of a foot*, the height of the connection point on the tree.